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Exploration of Autism Spectrum Disorder using Classification Algorithms

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Abstract

Data mining plays a vital role in identifying the classification algorithm for classification and prediction of disease in medical field. Predicting disease depends on the accuracy of the dataset and mechanism of machine learning algorithms used to classify the dataset. To detect new born with risk of autism spectrum disorder (ASD) which is predominantly diagnosing the behavioral observation of children with above 2 years. Although identifying brain disease using magnetic resonance imaging (MRI) available in the last few years it is time consuming and very expensive. Early stage of predicting autism disorder done with the help of using machine learning algorithms. This paper criticize various classification algorithms then evaluates the accuracy and then selects the best classification algorithm to analyze and classify each individual who gets affect from autism spectrum disorder (ASD) in early stage.

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Keywords: Data mining, Naive Bayes algorithm, Decision table, Confusion Matrix, Autism, Support Vector Machine.

1. Introduction

Data mining is the process of analyzing the large databases and extracting the hidden information from it using the methods of machine learning, statistics and data base system. It plays an important role in many fields like agriculture, medical, banking etc. This paper deals with data mining in medical field. There are several techniques

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available in data mining in which classification is one of the techniques used to assign items in a collection of target categories or classes [1]

Autism spectrum disorder is also called as ASD. The word Spectrum in ASD means that every child is unique and they have own characteristics. It talks about behavioral development issues affecting child in terms of communication, social, play skills and interest. ASD symptoms are not same for everyone. It may vary among people. It is classified into three different types.

The most typical types of disorders are Asperger syndrome and pervasive development. The training data set is taken from UCI repository. In this dataset, we use ten behavioral and individual characteristics that have been proved to be effective in detecting the ASD cases [4][2].

Main motive of developing algorithms based on subfield and mainly used for calculates accuracy, speed and customizability. The beneficitation of machine learning in medical for disease prediction and analyze the data. Among various algorithms most recurrently used algorithms are Naive Bayes, Decision table, and Support Vector machine.

Supervised learning is one of the machines learning function that maps the input and output. In this supervised learning algorithms are trained using training data set based on which predict the output value... The mostly used supervised learning algorithms are classification and regression. The output value either in the form of discrete or continuous value. Regression take place output values are continuous and classification take place output values are discrete [5].

2. Naïve Bayes Algorithms

Naive Bayes algorithm is a probabilistic classifier based on the Bayes theorem with assumption of independence among predictors. Bayes theorem describes the probability of occurrence of an event based on the related event's probability.

Naive Bayes is a model of conditional probability given a problem instance to be classified, represented by a vector $x = (x_1, x_2, \dots, x_n)$. The model makes an assumption that one particular feature in a class is not related to any other feature [6].

$$P(c_k | x_1, x_2, \dots, x_n) \quad (1)$$

$$\text{Posterior} = \frac{\text{Prior} \cdot \text{Probability}}{\text{Proof}} \quad (2)$$

Here the denominator is effectively constant. Naive Bayes is used for circumstances where all attributes are categorical value. As depicted in figure 1 in naïve Bayes algorithm the correctly classified instance shows that about 97.017 percent of classification has been done successfully and the remaining falls under the category of 2.93 percent.

Steps:

- The data set can be transferred to a frequency table.
- The Likelihood table is generated after finding the probabilities.
- Calculate the posterior probability for each class by using Naïve Bayesian equation.
- The class with the highest posterior probability is the outcome of Prediction.

2.1 Tables

Confusion matrix is a technique for summarize the performance of classification algorithms.

$$\text{Classification Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Number of Predictions}} \quad (3)$$

Classification accuracy can also be roll into a error rate by inverting the value, such as:

$$\text{Error Rate} = \left(1 - \left(\frac{\text{Correct Prediction}}{\text{Total Number of Predictions}} \right) \right) * 100 \quad (4)$$

The Table 1 represents the confusion matrix of Autism data set using Naïve Bayes algorithm in Weka Tool.

Table 1 Shows the Confusion matrix of Naive Bayes classifier Algorithm.

Total Number of instance N=704	Predicted No	Predicted Yes
Actual No	496	19
Actual Yes	2	187

3. Decision Table

Decision table is an exact method for predicting numeric values from the decision tree. It is more compact and understandable than decision trees [1][3]. Decision table is simple and computational time is less than the decision tree. In Weka tool, decision table algorithm is available under rules classifier. It summarizes the dataset with the decision table which contains the attributes which are same as the original data sets. The output shows the decision based on the number of attributes for each instance. The Table 2 represents the confusion matrix of Autism data set using decision tree algorithm in Weka Tool. As depicted in figure 2 the correctly classified instance shows that 100 percent of classification has been done successfully with Kappa statistics as 1 which implicates that the accuracy is achieved but not to the fullest.

3.1 Tables

Table 2 Shows the Confusion matrix of Decision Table

Total Number of instance N=704	Predicted No	Predicted Yes
Actual No	515	0
Actual Yes	0	189

4. Support Vector Machine

Support vector machine is a supervised learning algorithm. It can be used for both classification and regression. In Weka there are couple of implementations of support Vector machine. Under Function category SMO (Sequential Minimal Optimization) algorithm is implemented by Support vector machine classifier.

Support vector machine is a classification algorithm which takes training dataset as input and groups them to produce an optimal output. The Table 3 represents the confusion matrix of Autism data set using Support Vector Machine algorithm in Weka Tool. The representation of figure 3 shows the support vector machine algorithm produces 100 percent successful classification with kappa statistics as 1 that implicates that the accuracy is achieved to the fullest

4.1 Tables

Table 3 Shows the Confusion matrix of Support Vector Machine

Total Number of instance N=704	Predicted No	Predicted Yes
Actual No	515	0
Actual Yes	0	189

It reduces the classification errors and produces the best result compared to other two algorithms

5. Result and Discussion

The Table 4 represents some statistical value generated by Naïve Bayes, Decision Table and Support Vector Machine algorithm for Autism dataset in Weka Tool for the purpose of comparative analysis and the representation of figure 4,5,6 shows the cost benefit analysis of Naïve Bayes , Decision Table and Support Vector machine.

Table 4 Shows the Result Analysis of Three Algorithms

Error Rate Evaluation	Decision Table	Naive Bayes	SVM
Correctly classified instance	704	683	704
Incorrectly classified instance	0	21	0
Mean absolute Error	0.0031	0.326	0
Root mean squared error	0.0035	0.136	0
Relative absolute Error	0.7997	8.29	0
Root relative squared error	0.7998	30.69	0
Kappa statistic	1	0.9262	1

Compared to Decision Table, Naive Bayes and Support vector machine produces best result that means all the errors are reduced and kappa statistics shows that 100% data are reliable.

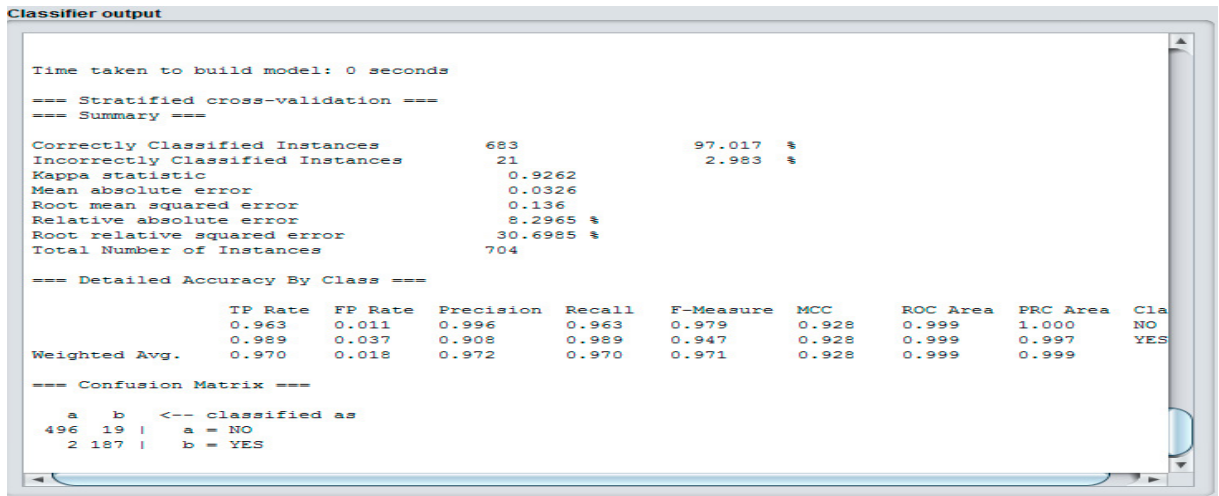


Fig. 1. Summary of Naive Bayes algorithm

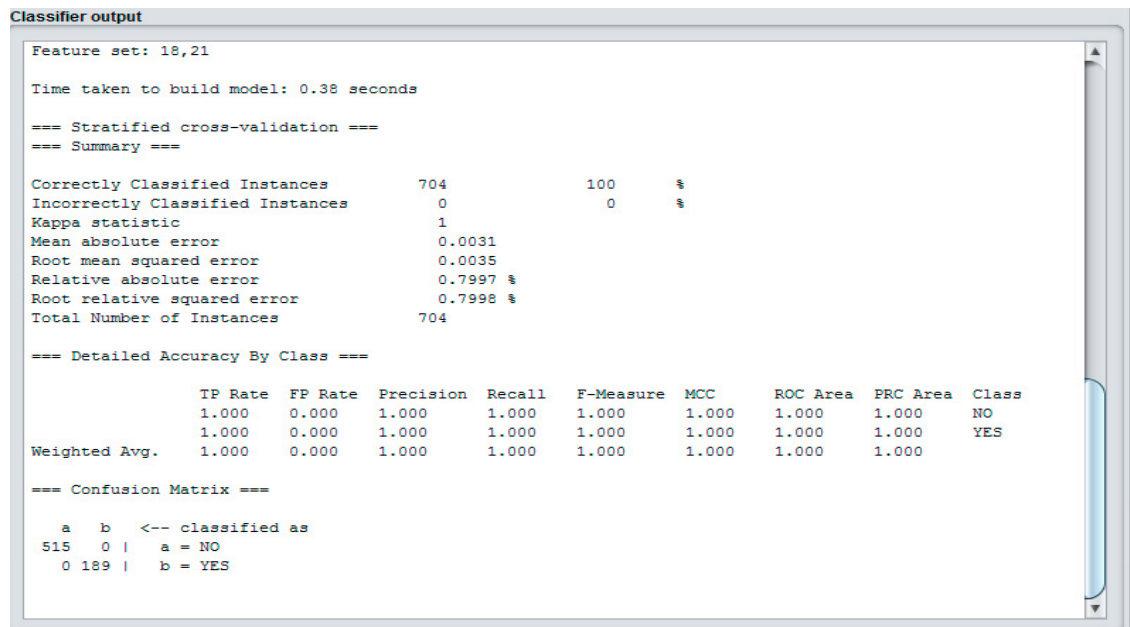


Fig. 2. Summary of Decision Table

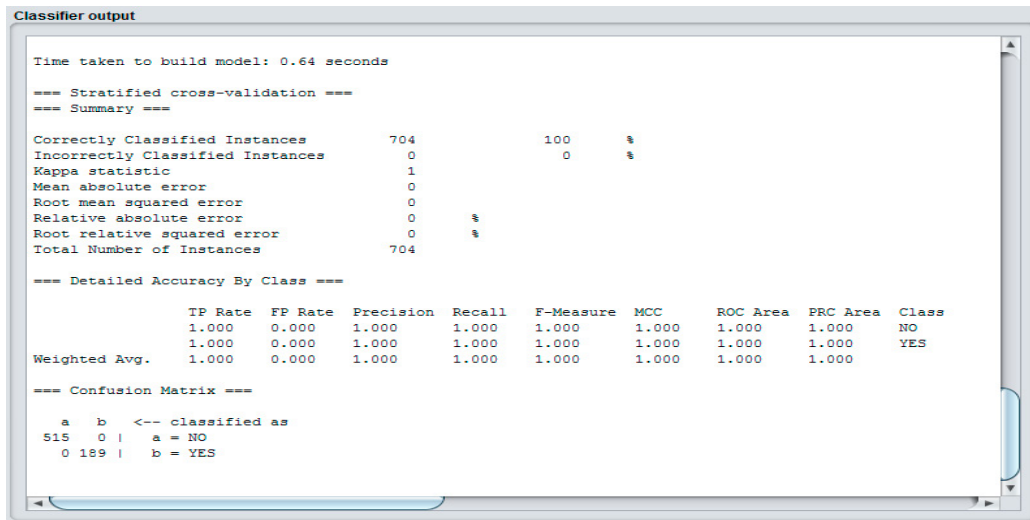


Fig 3: Summary of Support Vector Machine

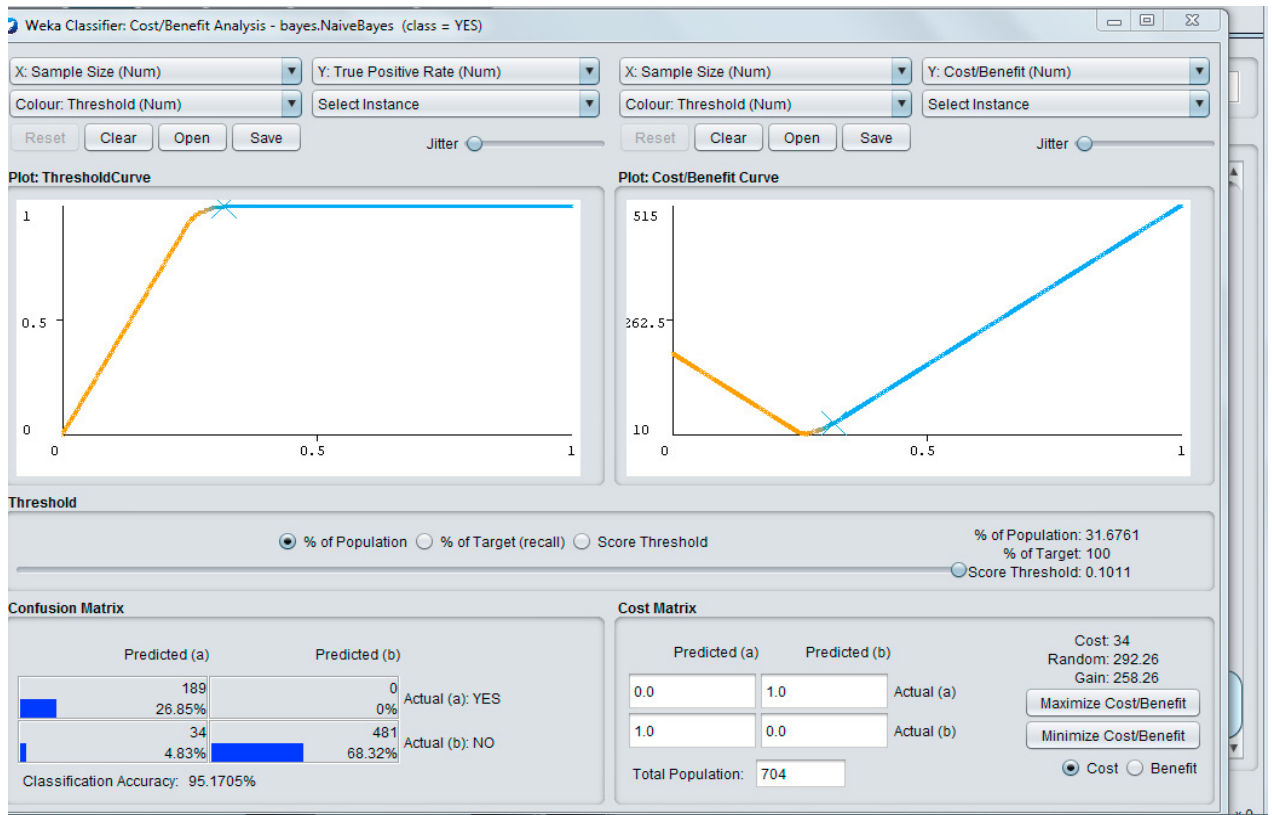


Fig. 4: cost benefit analysis of Naïve Bayes algorithm

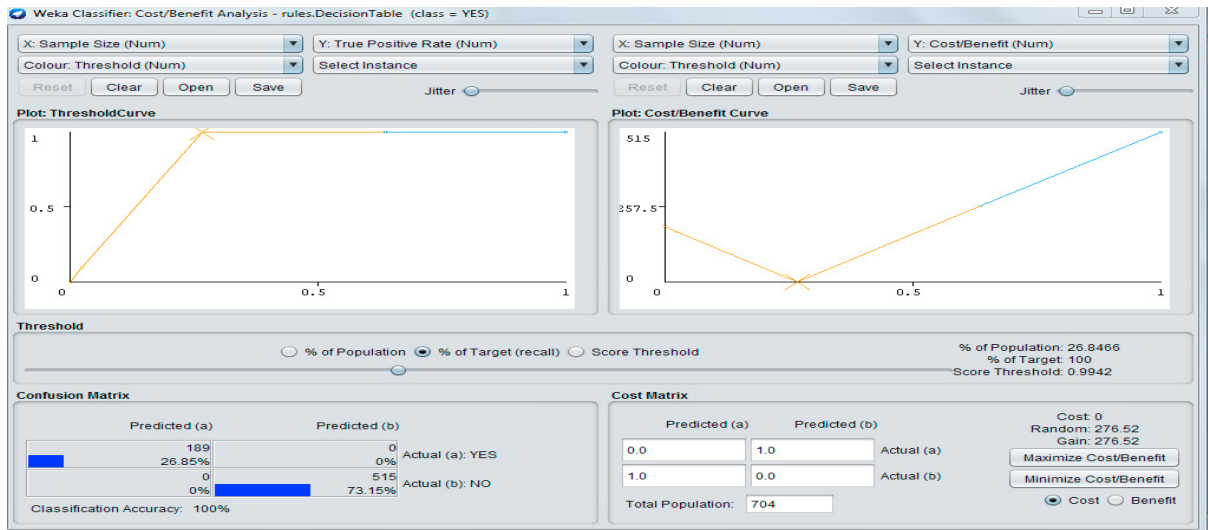


Fig. 5. Cost benefit analysis of Decision Table

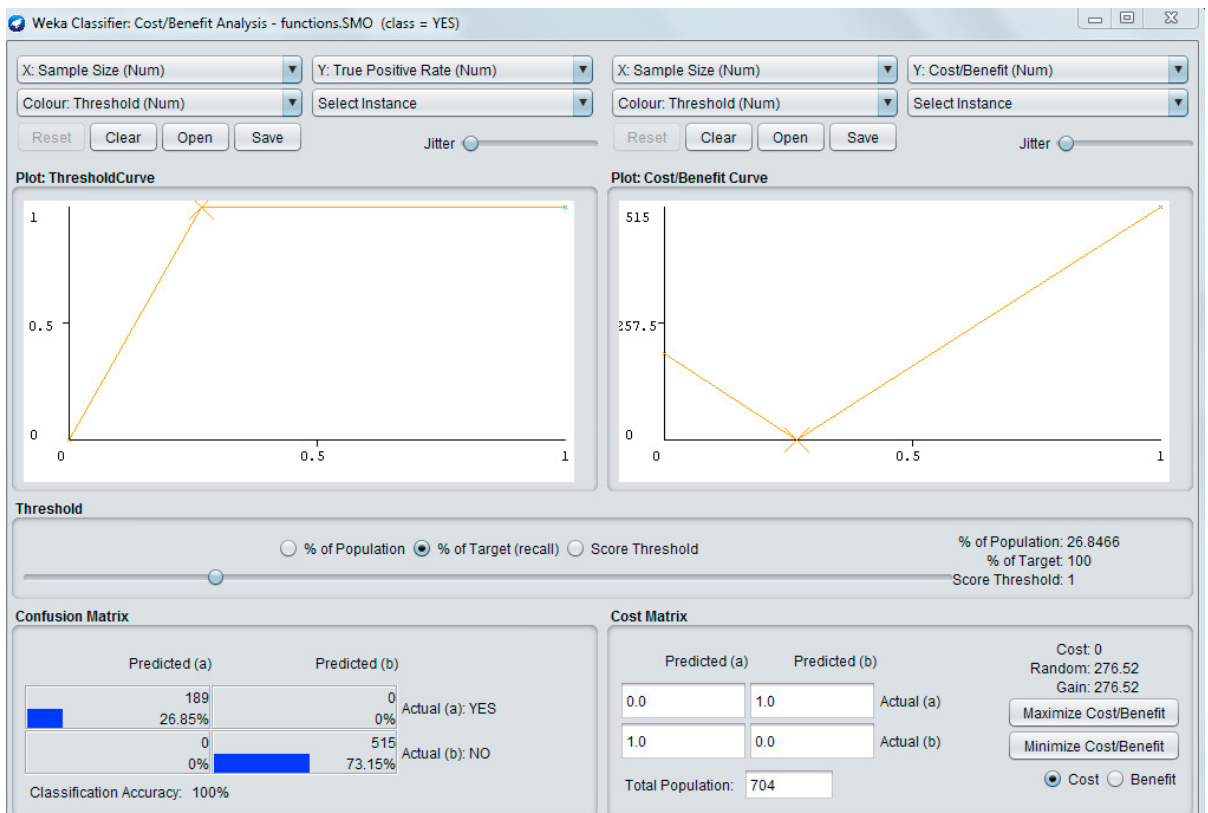


Fig. 6. Cost benefit analysis of SMO Algorithm using SVM Classifier

6. Conclusion

This paper focus to predict the autism spectrum disorder (ASD) in the best way to criticize three classification algorithms using training data set and find the outstanding Classifier. The Proposition of the paper is to perceive a machine learning prototype which are trained with set of attributes and produce outstanding result.

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